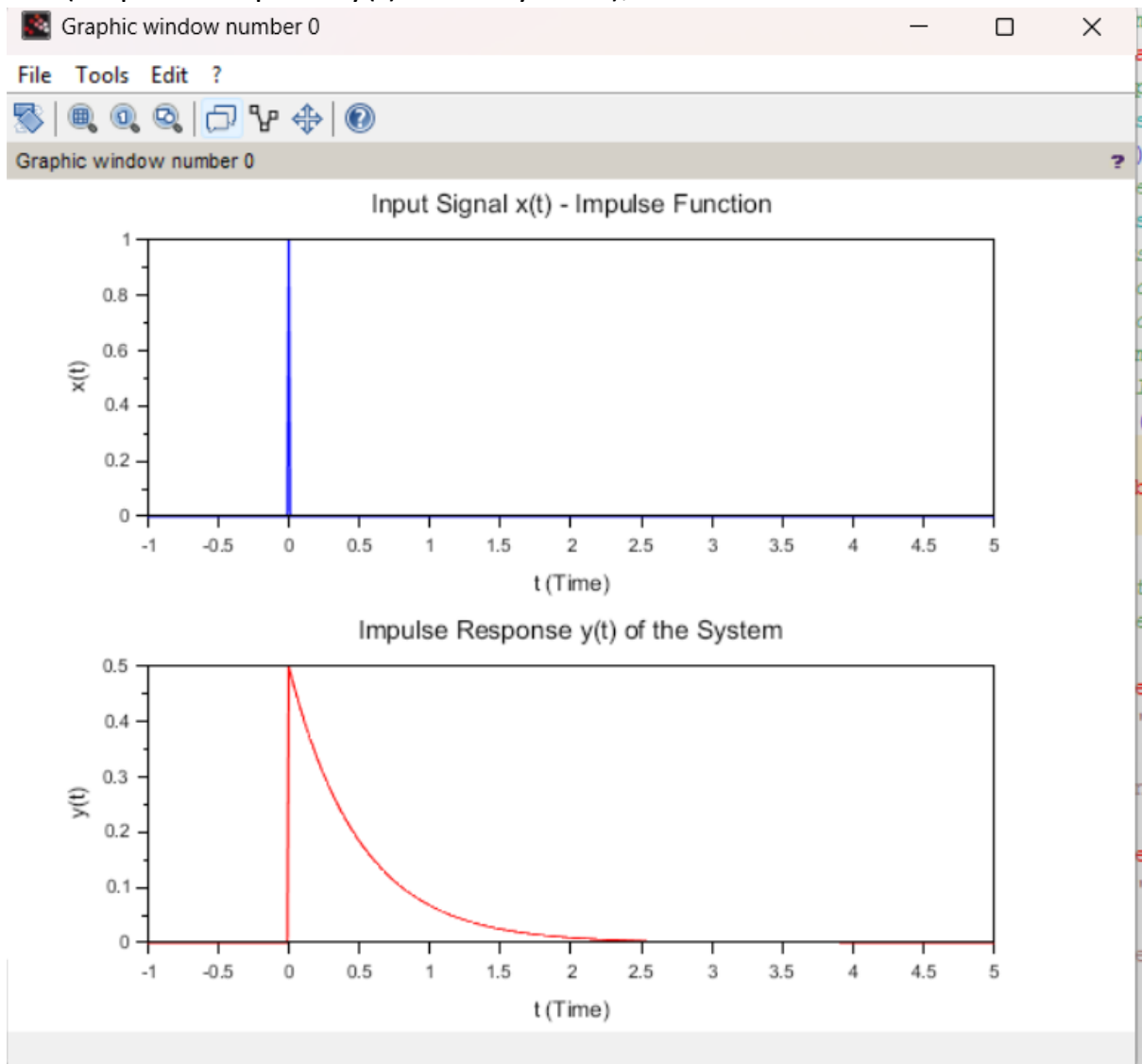


## 5. System Response and LTI Properties

### Experiment 5.1

```
// Test Case: Impulse Response of a Continuous-Time System
// Define time range and sampling interval
t_min = -1; // Start time
t_max = 5; // End time
dt = 0.01; // Time step
t = t_min:dt:t_max; // Time vector
// Define the impulse function
x_impulse = zeros(1, length(t));
x_impulse(t == 0) = 1; // Approximate impulse function
// Initialize the output signal with zeros
y_impulse = zeros(1, length(t));
// Define the system parameters
a = 2; // Coefficient for y(t)
b = 1; // Coefficient for x(t)
// Compute the impulse response using the Laplace Transform approach
// Analytical solution:  $y(t) = (b/a) * \exp(-a*t)$  for  $t \geq 0$ 
for i = 1:length(t)
    if t(i) >= 0
        y_impulse(i) = (b / a) * exp(-a * t(i));
    end
end
// Plot the input and output signals for impulse response
clf; // Clear the current figure
subplot(2,1,1);
```

```
plot(t, x_impulse, 'b');  
xlabel('t (Time)');  
ylabel('x(t)');  
title('Input Signal x(t) - Impulse Function');  
subplot(2,1,2);  
plot(t, y_impulse, 'r');  
xlabel('t (Time)');  
ylabel('y(t)');  
title('Impulse Response y(t) of the System');
```



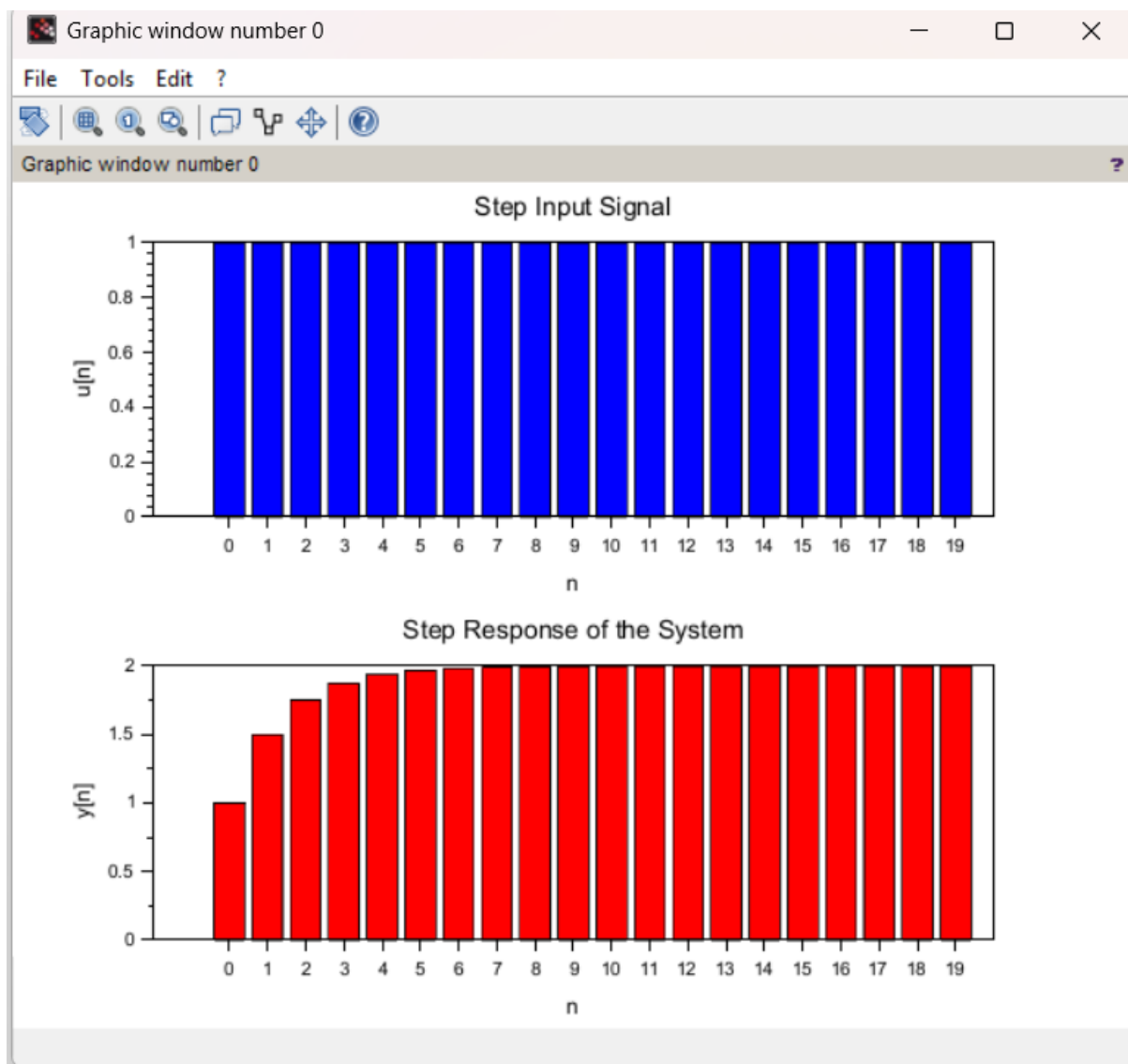
## Experiment 5.2

```
// Test Case: Step Response of a Discrete-Time System
// Define the length of the signal
N = 20; // Length of the signal
// Define the input signal (step function)
u_n = ones(1, N); // Step function
// Define the system parameters
b = [1]; // Numerator coefficients of the system (input)
a = [1, -0.5]; // Denominator coefficients of the system (output)
// Compute the system's response to the step input
y_n = filter(b, a, u_n); // System output for step input
// Define the time index for the signals
n = 0:N-1; // Time index
// Plot the step input and system response
clf; // Clear the current figure

// Plot step input
subplot(2,1,1);
bar(n, u_n, 'b');
xlabel('n');
ylabel('u[n]');
title('Step Input Signal');
// Plot system response to step input
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n');
```

```
ylabel('y[n]');
```

```
title('Step Response of the System');
```



### Experiment 5.3

```
h = [1 2 1];
```

```
x1 = [1 2];
```

```
x2 = [3 4];
```

```
a = 2; b = -1;
```

```
y1 = conv(x1, h);
```

```
y2 = conv(x2, h);
```

```
lhs = a*y1 + b*y2;
```

```
rhs = conv(a*x1 + b*x2, h);
```

```
disp("LHS: "); disp(lhs);
```

```
disp("RHS: "); disp(rhs);
```

```
"LHS: "  
-1.  -2.  -1.   0.  
"RHS: "  
-1.  -2.  -1.   0.
```

```
_ \ |
```

### Experiment 5.4

```
// Test Case: Analysis of a Discrete-Time System
```

```
// Define the input signal (unit impulse)
```

```
N = 20; // Length of the signal
```

```
x_n = zeros(1, N); // Initialize input signal
```

```
x_n(1) = 1; // Unit impulse at n=0
```

```
// Define the system parameters
```

```
b = [1]; // Numerator coefficients of the system (input)
```

```
a = [1, -0.5]; // Denominator coefficients of the system (output)
```

```
// Compute the impulse response using the filter function
```

```
h_n = filter(b, a, x_n); // Impulse response of the system
```

```
// Define a different input signal (e.g., a step function)
```

```
u_n = ones(1, N); // Step function
```

```
// Compute the system's response to the step input
```

```
y_n = filter(b, a, u_n); // System output for step input
```

```
// Define the time index for the signals
```

```
n = 0:N-1; // Time index
```

```
// Plot the impulse response and system response
```

```
clf; // Clear the current figure
// Plot impulse response
subplot(2,1,1);
bar(n, h_n, 'b');
xlabel('n');
ylabel('h[n]');
title('Impulse Response of the System');
// Plot system response to step input
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n');
ylabel('y[n]');
title('Response of the System to Step Input');
```

